

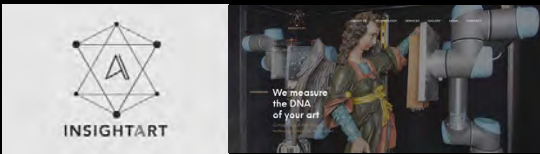
As part of the research, detailed photo-documentation was carried out, including macro and micro images, photographs with raking light, and verification images of *Infrared Reflectography* and *Ultraviolet Luminescence*. This work also includes X-ray images using scanning technology drawn from CERN technology and used by NASA. The technology provides pseudo-color X-ray scans, revealing differences in material compositions. Results from previous scans and testing are compared and commented upon in the complete report.

Signature - Radiographic Analysis and Comparison:

The signature was extensively examined for the purpose of its authenticity and belonging to the layers of painting itself. A survey of the material composition of the individual letters of the signature was carried out, which confirmed the presence of ivory black. The possibility of the occurrence of bone black with respect to trace amounts of other detected elements is also mentioned. Furthermore, the method of craquelure through individual letters of the signature was examined using a microscope.

A radiographic analysis of the signatures was also performed and its comparison with other signatures from Modigliani's well-known works. See 2022 letter to the founder of InsightART, Jiri Lauterkranc and their Testing Report. The results of this survey show that there is a strong agreement on the basic characteristics of Modigliani's signature style. Based on these aspects, and similar data from 2011 and 2014 it can be stated that the signature is genuine.

2022 Forensic Report - InsightART - Excerpt



Sitting Nude with Hands Crossed

InsightART

Jiří Lauterkranc PhD

Prague

2022

Title: Sitting Nude with Hands Crossed
Author: Modigliani
Date: 1917
Signature: Yes
Medium: Oil on Canvas
Dimensions: 40.5 x 33 cm
Source: Private Coll. Prague
Owner: Upon Inquiry/ NDA



Description

The painting being examined depicts a seated female nude sitting on a reddish pillow with greenish tassel trim in a darkened room. The seated female figure has her legs crossed and both arms joined below her left knee. The whole scene is set in the interior of the room, which is painted in dark shades of brown, ochre and green. A still life with a table and chair is painted in the upper right corner. There is a partial white tablecloth visible. On the left side lower edge is a reddish-brown drape. In the upper left side leading into the top center is what appears to be a wooden cabinet with a bowl and a gray looking cover on top of the cabinet. In the lower right corner is the signature "modigliani" in black paint.

Research Methodology

This examination was carried out based on requirement to verify the authenticity and style of the artwork in terms of technological and artistic quality. This work of art has already been the subject of several exhaustive technological analyses in the past (2011-Rome, 2014-Florence – see Exhibits), focusing mainly on its material composition, technical condition and subsequent restoration interventions. The subject of this study is therefore also a comparison of all previous results and their complementation with new findings and analyses. This work also includes a comparison of the artist's painting style, technique, materials, and signature. As part of the research, detailed photo-documentation of the front and back of the painting in visible daylight was carried out, as well as macro and micro images and photographs in bright side lighting (raking light). A verification image of Infrared Reflectography and Ultraviolet Luminescence was also taken.

This work also includes X-Ray images using InsightART's RToo scanning technology. Drawn from CERN technology and used by NASA, the RToo provides pseudo-color X-ray scans, revealing differences in material compositions. Additional information is included in exhibits to this report. Results from previous scans and testing are compared and commented upon in this report.

Exhibit-5a
ARSMensurae - 2011 Pigment Test

Chart
The locations of test point numbers (column 1) are shown on the image in Exhibit-6b

ARSMENSVRAE

N° Point	color	Chemical elements	Proposed pigments
1	pink	Pb+++ , Zn, Fe, Hg	Biacca, white zinc, cinnabar
2	red	Pb+++ , Zn, Fe+, Cd+,	Biacca, white zinc , Cadmium pigment
3	white	Pb+++ , Zn, Fe,	Biacca, white zinc
4	red	Pb+++ , Zn, Fe+, Cd+,	Biacca, white zinc, ochre, Cadmium pigment
5	red	Pb+++ , Zn, Fe+, Cd,	Biacca, white zinc, ochre,
7	dark green	Pb+++ , Zn, Fe+, Cd++, Co+	Biacca, white zinc, Ochre, cobalt pigment, Cadmium pigment
7	dark green	Pb+++ , Zn, Fe+, Cd++, Co+	Biacca, white zinc, Ochre, cobalt pigment, Cadmium pigment
8	green	Pb+++ , Zn, Fe+, Co	Biacca, white zinc, Ochre, cobalt pigment
9	pink	Pb+++ , Zn, Fe, ,	Biacca, white zinc, ochre
10	Dark red	Pb+++ , Zn, Fe++, Cd, Hg	Biacca, white zinc, ochre, Cadmium pigment, cinnabar
11	gray	Pb+++ , Zn, Fe,	Biacca, white zinc, ochre
12	signature	Pb+++ , Zn, Fe+,	Biacca, white zinc, Ochre, cobalt pigment, Cadmium pigment

Exhibit-5b

ART-Test s.a.s. - 2014 Pigment Test Chart

The locations of test point numbers (column 1) are shown on the image in

Exhibit-6a

Table 1: Description of the XRF measuring points and the corresponding constituent materials.

Point	Colour	Description	Identified Materials
P1	Brownish red	Draperies	Pigments based on iron oxides, darkened by the addition of cobalt blue and green chromium, mixed and/or overlapped with lead white
P2	Blue	Highlighting on the pillow	Cobalt Blue lying above the layer consists of red-brown pigments based on iron oxide, cadmium red and cinnabar, mixed and/or overlapped with lead white
P3	Red	Pillow	Pigments based on iron oxide, cadmium red and cinnabar, mixed and/or overlapped with lead white
P4	Brownish green	Background	Pigments based on oxides of iron and manganese darkened with cobalt blue and chrome green, mixed and/or overlapped with lead white
P5	Skin tones	Arm characterized by an intense UV fluorescence	Zinc white and pigments based on iron oxides, mixed and/or overlapped with lead white
P6	Skin tones	Shoulder	Lead white in mixture with cinnabar and pigments based on iron oxides, added in low contents to obtain the skin tones tones
P7	Skin tones	Breast, characterized by an intense UV fluorescence	Zinc white, darkened adding cobalt blue and pigments based on iron oxide, mixed and/or overlapped with lead white
P8	Skin tones	Leg	Lead white in mixture with cinnabar and pigments based on iron oxides added in low contents in order to obtain skin tones tones.
P9	Skin tones (brownish red)	Left arm	Lead white mixed with pigments based on iron oxides added to obtain reddish brown tone of the skin.
P10	White	Table cloth	Lead white in overlap with pigments based on iron oxides
P11	Green	Background	In addition to pigments based on iron oxide and chromium green, the presence of cadmium red and cobalt blue has been verified in correspondence with this area. The lead white is to be referred to the preparation.
P12	Black layer characterized by blue shades	Hair	Black layer unidentifiable by XRF characterized by blue shades of cobalt blue. The lead white is to be referred to the preparation.
P13		Eyebrows	
P14	Bright red	Lips	Pigments based on iron oxides, cadmium red and cinnabar, mixed and/or overlapped with lead white. The darker layer is characterized by a higher iron signal, and contextually lowest for mercury (cinnabar), cadmium and selenium (cadmium red).
P15	Dark red		
P16	Brown	Background (near at the signature)	Pigments based on iron oxides, darkened with cobalt blue, mixed and/or overlapped with lead white.
P17	Black	Signature	Bone black or ivory black (hypothetised the higher content of calcium and for the presence of phosphorus) in overlapping to the composition identified for the point P16.
P18			

The preparation layer has been identified to lead white. The red layer showed different mixtures consisting of pigments based on of iron and manganese oxides, cadmium red and cinnabar.

The black and blue layers are characterized by the presence of cobalt to be correlated to the use of cobalt blue.

The pictorial layer characterized by a strong fluorescence in UV were identified as zinc white. This pigment is present with a very low intensity signal in the other areas realized with lead white (and pigments based on iron oxides and cinnabar in quantities necessary to achieve the desired shade).

Regarding the "Modigliani" signature was found to have a higher content of phosphorus and calcium, which allow to suppose the use of black ivory (or bone black).

Exhibit-5c
Rustin Levenson-Pigment Research
Chart-2000 Publication: "Seeing
Through Paintings"

Pigment Identification

Because the composition and availability of artists' pigments are well documented, the identification of certain pigments with known dates of introduction and geographical distribution can help confirm or dispute a painting's identification. The following table gives art historians a general idea of the most common pigments found during certain historical spans. Unless noted, pigments have no terminal date of use, although artists tended to favor more modern substitutes that were cheaper or more stable. Virtually all pigments known from ancient times have been commercially produced in the twentieth century. Therefore, pigment

studies alone can confirm that paint *cannot* be old but cannot confirm that it definitely is old.

The only counterexample is the presence of lead-tin yellow—a pigment used from at least the fourteenth century that inexplicably fell out of use in about 1750. Lead-tin yellow was rediscovered in 1941; hence it does not appear either in nineteenth-century paintings or in fakes produced prior to its rediscovery, such as Van Meegeren's "Vermeers." For an example of its significance, see Case Study 16. The definitive discussion is Herman Kühn's article in *Artists' Pigments 2*, pp. 83–102.

Van Meegeren was careful enough to use natural ultramarine (lapis lazuli) in his fakes; the faker was caught by the fraudulence of his supplier. Van Meegeren's paint manufacturer had adulterated the ultramarine with cobalt blue, a nineteenth-century introduction.

For more information, see Otto Kurz, *Fakes* (1948; reprint, New York: Dover, 1967), pp. 329–34.

A number of the works listed in the Bibliography include laboratory studies of pigments in the oeuvre of a particular artist or school; these would be the best starting points for scholars interested in pigment use by period, region, or artist. Gettens and Stout 1966 is an accessible source for a broad range of pigments, including those not mentioned in the table. *Artists' Pigments* has the most current and complete discussion of analytic techniques. The chapter on each pigment includes a list of notable occurrences; these should be very useful to art historians.

Chronology of Common Pigments According to Initial Use

	Ancient Times	Renaissance	18th Century	19th Century	20th Century
White	chalk gypsum lead white			zinc white	titanium white
Black	charcoal lamp black bone black				
Red	iron oxide reds (red ochres) red lead cinnabar (vermilion) red lakes (kermes, lac, madder)	red lake (carmine)		red lakes (synthetic)	cadmium red
Yellow	yellow ochre orpiment	Naples yellow <i>used approx. 1600–1800</i> lead-tin yellow <i>used approx. 1300–1750</i>		chrome yellow cadmium yellow cobalt yellow	zinc yellow
Green	malachite verdigris terre verte	copper resinsates <i>mostly 15th–16th cent.</i>	Scheel's green	cobalt green emerald green chrome green	
Blue	azurite indigo ultramarine <i>mostly 14th–15th cent.</i>	smalt <i>used late 15th–19th cent.</i>	Prussian blue	cobalt blue syn. ultramarine cerulean blue	manganese blue

Exhibit-6a

ART-Test s.a.s. - Forensic Scan Points

X-ray fluorescence was performed on 20 points, analyzed to identify the pigments in the pictorial layer in order to obtain useful information for the authentication study and indirectly dating of the painting. In order to guarantee the significance outcomes, the measurement points have been selected on the basis of information provided by the multispectral/multimodal imaging investigations, so to avoid restoration areas.

Through the acquisition of the spectra on the 20 measurement points, localized in figure 27, the painted layers on the following figurative elements have been studied: body and face of the woman (skin tones, lips), details of the internal environment, brown background near the signature, the signature, and the preparation layer (visible along the side edge), back (canvas).

All 20 points are included in the attached testing report with points 19 & 20 on the canvas back.

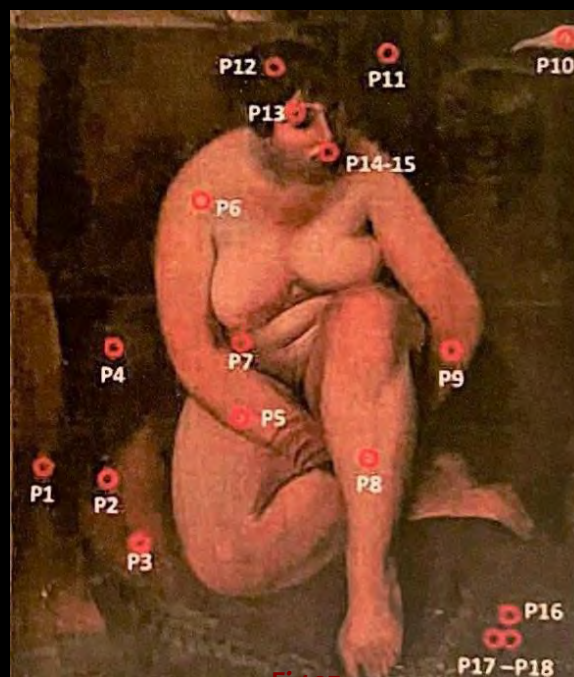
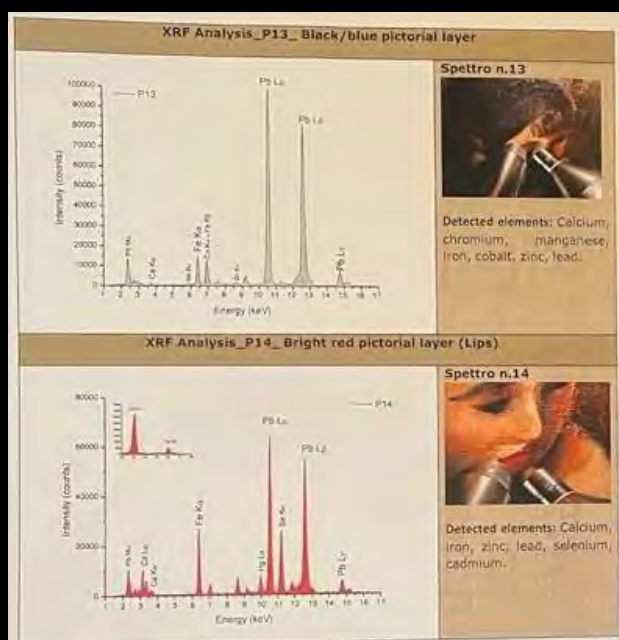


Fig 27

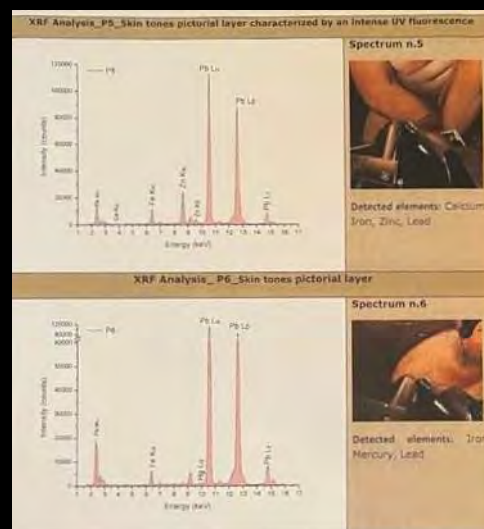
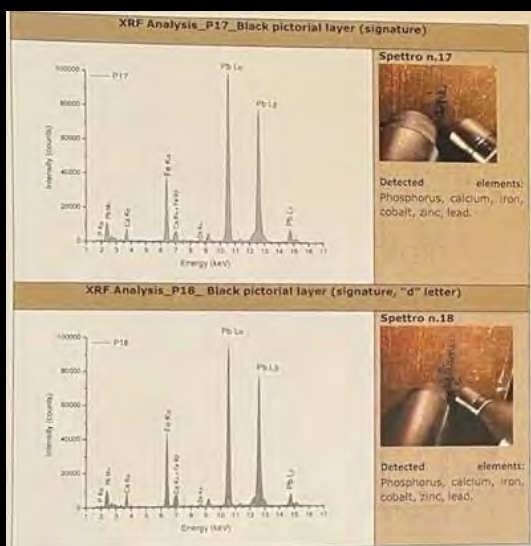


Exhibit-6b
ARS
Mensvrae - Forensic Scan Points

The painting has been analyzed in 12 point availing of X ray fluorescence (EDXRF). In Figure 51 it is possible to see the analyzed spots. Apart from the spots on the verso of the painting, one single measurement was fulfilled on the preparation of the canvas.

All 12 points are included in the attached testing report.



12 Points of Pigment Testing – see chart in 5a



Micro-photography Detail of Face.

In all the measures there is the presence of white lead (biacca), white zinc and ochre, these elements are connected to the preparation of the canvas. Ochre's and Terre are widely used in the painting. The red pigment is cadmium red, while the blue is cobalt blue. It is documented the use of cinnabar to make the red more dark.

In the next figures we report some of the XRF spectrum results.

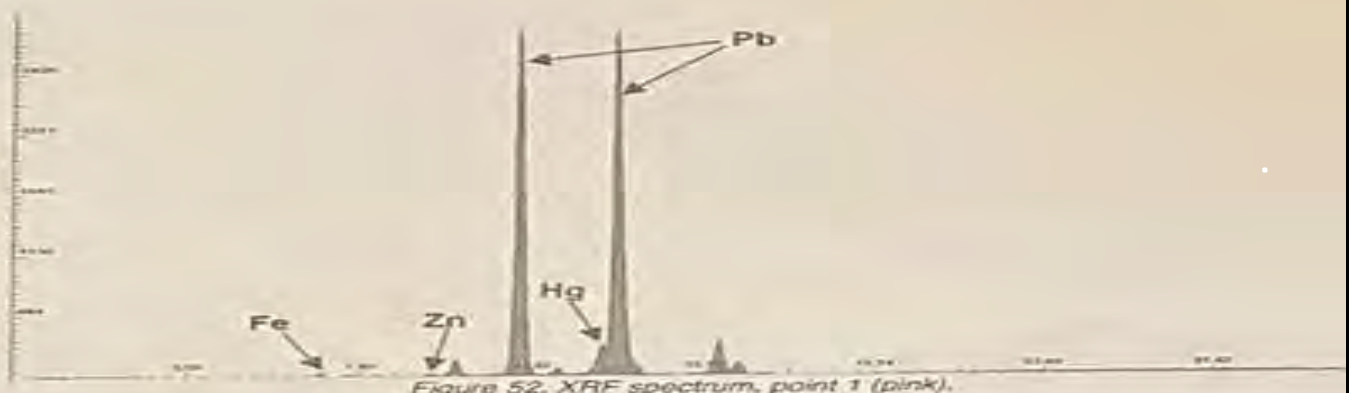


Figure 52. XRF spectrum, point 1 (pink).

Exhibit 11 from Authentication & Analysis Report

Confirming Modigliani Painting Style & Unique Techniques

The most interesting discovery followed an investigation of the under-painting, i. e. how the artist prepared his composition. They discovered that details of the face, nose, mouth, and eyes were uniformly drawn in “large and violent” rings of brown paint. These circular brushworks were so unvarying that this was a conclusive way to identify Modigliani’s hand. Genuine works always showed this artistic signature; faked works never did.

The reader can see these brown circles are on this painting! That is as good as the signature of Modigliani being confirmed by three testing services as original and in “bone black” pigment and signed at the painting completion. Other details supporting this painting as a “style” by Modigliani are the lines and the body parts as discussed in items 9,10, and 11. This is yet another “atypical, but genuine” painting by Modigliani, similar to the Diego Rivera that was confirmed using the same analysis data.

Excerpts from Meryle Secrest 2011 Book “Modigliani A Life” pages 330 & 331



Amedeo Modigliani

*"When I know your soul,
I will paint your eyes"*

